

HDOC DAY : 24

AIM: To write a C program that displays a menu to print different increasing patterns based on the user's choice using a switch statement, nested loops, and parameterless functions with no return value.

ALGORITHM:

- 1. Start**
2. Define five functions: pattern1(), pattern2(), pattern3(), pattern4(), and pattern5().
3. In main():
 - a. Display a menu listing the 5 pattern choices.
 - b. Ask the user to enter their choice (1 to 5).
 - c. Use a switch statement to call the corresponding function based on user input.
4. Inside each pattern function:
 - a. Read the number of rows n from the user.
 - b. Use nested loops (for loops) to print the specific pattern row by row.
- 5. Stop**

TEST CASE TABLE -

TEST CASE NO	Selected option	Input (n)	Expected output
1	1. Right-angled triangle	3	<pre>* **</pre>

2	2. Row-number triangle	3	1 22 333
3	3. Natural-number pattern	3	1 12 123
4	4. Consecutive-number triangle	3	1 2 3 4 5 6
5	5. Alphabet pattern	3	A AB ABC

C CODE -

```
#include <stdio.h>
```

```
void pattern1() {  
    int n, i, j;  
    printf("Enter number of rows: ");  
    scanf("%d", &n);  
    for(i = 1; i <= n; i++) {  
        for(j = 1; j <= i; j++) {  
            printf("*");  
        }  
        printf("\n");  
    }  
}
```

```
void pattern2() {  
    int n, i, j;  
    printf("Enter number of rows: ");  
    scanf("%d", &n);  
    for(i = 1; i <= n; i++) {  
        for(j = 1; j <= i; j++) {  
            printf("%d", i);  
        }  
    }  
}
```

```
    }  
    printf("\n");  
}  
}
```

```
void pattern3() {  
    int n, i, j;  
    printf("Enter number of rows: ");  
    scanf("%d", &n);  
    for(i = 1; i <= n; i++) {  
        for(j = 1; j <= i; j++) {  
            printf("%d", j);  
        }  
        printf("\n");  
    }  
}
```

```
void pattern4() {  
    int n, i, j, count = 1;  
    printf("Enter number of rows: ");  
    scanf("%d", &n);  
    for(i = 1; i <= n; i++) {  
        for(j = 1; j <= i; j++) {
```

```
        printf("%d ", count);  
        count++;  
    }  
    printf("\n");  
}  
}
```

```
void pattern5() {  
    int n, i, j;  
    printf("Enter number of rows: ");  
    scanf("%d", &n);  
    for(i = 1; i <= n; i++) {  
        for(j = 1; j <= i; j++) {  
            printf("%c", 'A' + j - 1);  
        }  
        printf("\n");  
    }  
}
```

```
int main() {  
    int choice;  
    printf("--- PATTERN MENU ---\n");  
    printf("1. Increasing right-angled triangle\n");
```

```
printf("2. Increasing row-number triangle\n");  
printf("3. Increasing natural-number pattern\n");  
printf("4. Consecutive-number triangle\n");  
printf("5. Increasing alphabet pattern\n");  
printf("Enter your choice (1-5): ");  
scanf("%d", &choice);
```

```
switch(choice) {
```

```
    case 1:
```

```
        pattern1();
```

```
        break;
```

```
    case 2:
```

```
        pattern2();
```

```
        break;
```

```
    case 3:
```

```
        pattern3();
```

```
        break;
```

```
    case 4:
```

```
        pattern4();
```

```
        break;
```

```
    case 5:
```

```
        pattern5();
```

```
        break;
```

default:

```
printf("Invalid choice!\n");
```

```
}
```

```
return 0;
```

```
}
```

CODE SCRRENSHOT :

```
#include <stdio.h>

void pattern1() {
    int n, i, j;
    printf("Enter number of rows: ");
    scanf("%d", &n);
    for(i = 1; i <= n; i++) {
        for(j = 1; j <= i; j++) {
            printf("*");
        }
        printf("\n");
    }
}

void pattern2() {
    int n, i, j;
    printf("Enter number of rows: ");
    scanf("%d", &n);
    for(i = 1; i <= n; i++) {
        for(j = 1; j <= i; j++) {
            printf("%d", i);
        }
        printf("\n");
    }
}

void pattern3() {
    int n, i, j;
    printf("Enter number of rows: ");
    scanf("%d", &n);
    for(i = 1; i <= n; i++) {
        for(j = 1; j <= i; j++) {
            printf("%d", j);
        }
        printf("\n");
    }
}

void pattern4() {
    int n, i, j, count = 1;
    printf("Enter number of rows: ");
    scanf("%d", &n);
    for(i = 1; i <= n; i++) {
        for(j = 1; j <= i; j++) {
            printf("%d ", count);
            count++;
        }
        printf("\n");
    }
}

void pattern5() {
    int n, i, j;
    printf("Enter number of rows: ");
    scanf("%d", &n);
```

```

        for(i = 1; i <= n; i++) {
            for(j = 1; j <= i; j++) {
                printf("%d ", count);
                count++;
            }
            printf("\n");
        }
    }

void pattern5() {
    int n, i, j;
    printf("Enter number of rows: ");
    scanf("%d", &n);
    for(i = 1; i <= n; i++) {
        for(j = 1; j <= i; j++) {
            printf("%c", 'A' + j - 1);
        }
        printf("\n");
    }
}

int main() {
    int choice;
    printf("---- PATTERN MENU ----\n");
    printf("1. Increasing right-angled triangle\n");
    printf("2. Increasing row-number triangle\n");
    printf("3. Increasing natural-number pattern\n");
    printf("4. Consecutive-number triangle\n");
    printf("5. Increasing alphabet pattern\n");
    printf("Enter your choice (1-5): ");
    scanf("%d", &choice);

```

```
switch(choice) {  
    case 1:  
        pattern1();  
        break;  
    case 2:  
        pattern2();  
        break;  
    case 3:  
        pattern3();  
        break;  
    case 4:  
        pattern4();  
        break;  
    case 5:  
        pattern5();  
        break;  
    default:  
        printf("Invalid choice!\n");  
}  
  
return 0;  
}
```

OUTPUT SCREENSHOT :

```
[manalithakar@Unknown_4e:f1:c3:0c:fe:82 ~ % nano 24.c
[manalithakar@Unknown_4e:f1:c3:0c:fe:82 ~ % gcc 24.c -o 24
[manalithakar@Unknown_4e:f1:c3:0c:fe:82 ~ % ./24
--- PATTERN MENU ---
1. Increasing right-angled triangle
2. Increasing row-number triangle
3. Increasing natural-number pattern
4. Consecutive-number triangle
5. Increasing alphabet pattern
Enter your choice (1-5): 1
Enter number of rows: 3
*
**
***
[manalithakar@Unknown_4e:f1:c3:0c:fe:82 ~ % ./24
--- PATTERN MENU ---
1. Increasing right-angled triangle
2. Increasing row-number triangle
3. Increasing natural-number pattern
4. Consecutive-number triangle
5. Increasing alphabet pattern
Enter your choice (1-5): 2
Enter number of rows: 3
1
22
333
[manalithakar@Unknown_4e:f1:c3:0c:fe:82 ~ % ./24
--- PATTERN MENU ---
1. Increasing right-angled triangle
2. Increasing row-number triangle
3. Increasing natural-number pattern
4. Consecutive-number triangle
5. Increasing alphabet pattern
Enter your choice (1-5): 3
Enter number of rows: 3
1
12
123
```

manalithakar@Unknown_4e:f1:c3:0c:fe:82 ~ % ./24

--- PATTERN MENU ---

1. Increasing right-angled triangle
2. Increasing row-number triangle
3. Increasing natural-number pattern
4. Consecutive-number triangle
5. Increasing alphabet pattern

Enter your choice (1-5): 4

Enter number of rows: 3

1

2 3

4 5 6

manalithakar@Unknown_4e:f1:c3:0c:fe:82 ~ % ./24

--- PATTERN MENU ---

1. Increasing right-angled triangle
2. Increasing row-number triangle
3. Increasing natural-number pattern
4. Consecutive-number triangle
5. Increasing alphabet pattern

Enter your choice (1-5): 5

Enter number of rows: 3

A

AB

ABC

manalithakar@Unknown_4e:f1:c3:0c:fe:82 ~ % █